OTTO VON GUERICKE UNIVERSITÄT MAGDEBURG

MB FAKULTÄT FÜR MASCHINENBAU

IWF

Current research at the Chair of Metallic Materials

December, 2024
with some updates

Group of Prof. Dr.-Ing. habil.Thorsten Halle
Otto-von-Guericke University Magdeburg,
Chair of Metallic Materials

thorsten.halle@ovgu.de

Who we are

Chair of Metallic Materials at the Institute of Materials, Technologies and Mechanics



Prof. Dr.-Ing. habil
Thorsten Halle
Chair

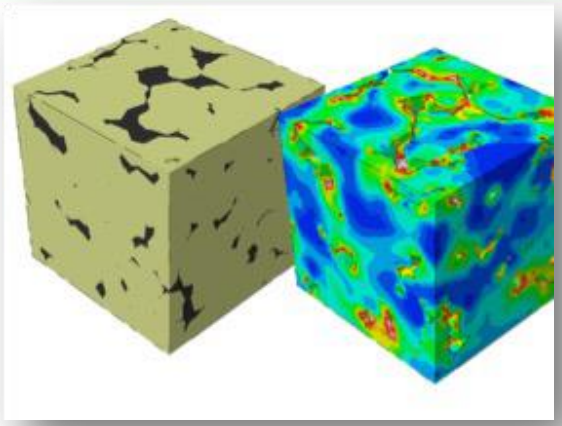


Dr.-Ing.
Hanka Becker
Emmy Noether Group
„New Alloys Made from Recycled Aluminum”



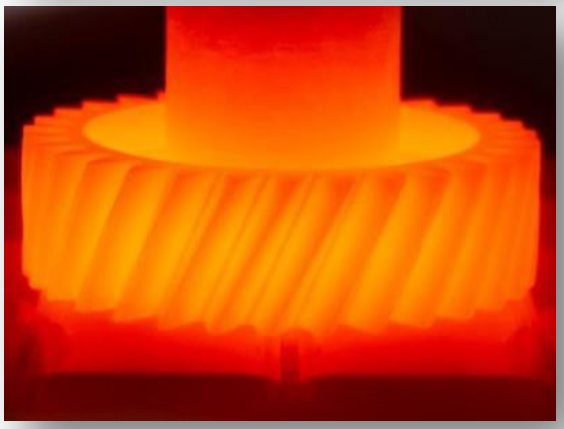
Dr.-Ing.
Sebastian Hütter
Group Lead
Computational Materials Science



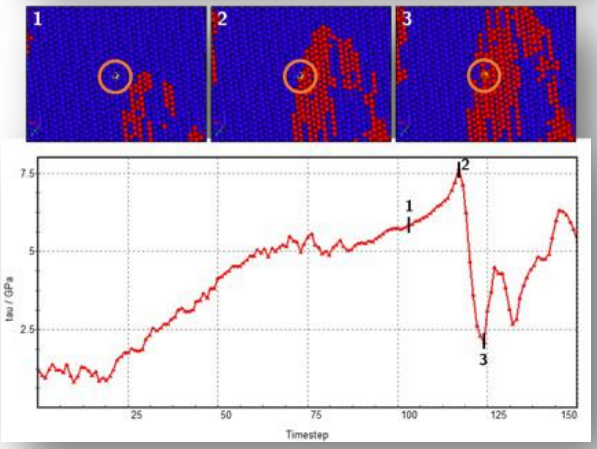


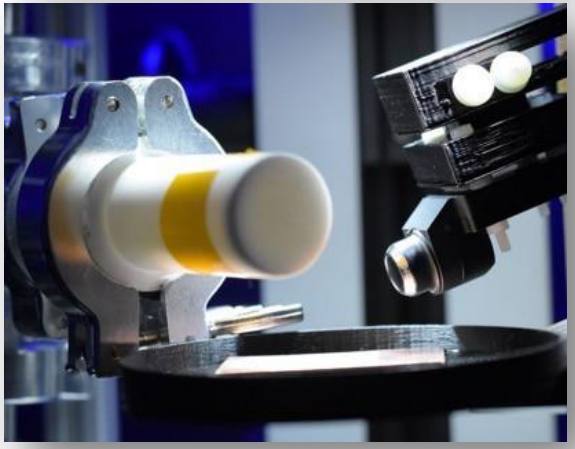


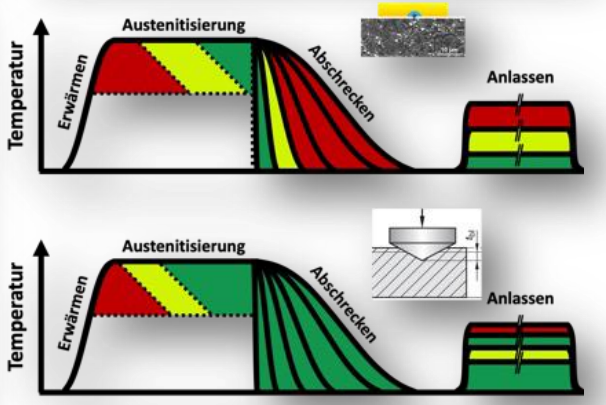

10 µm



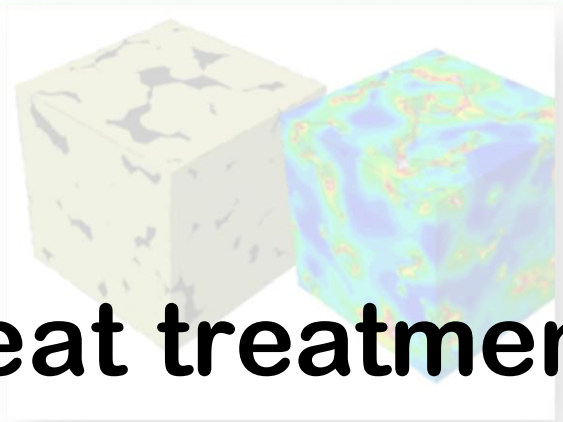


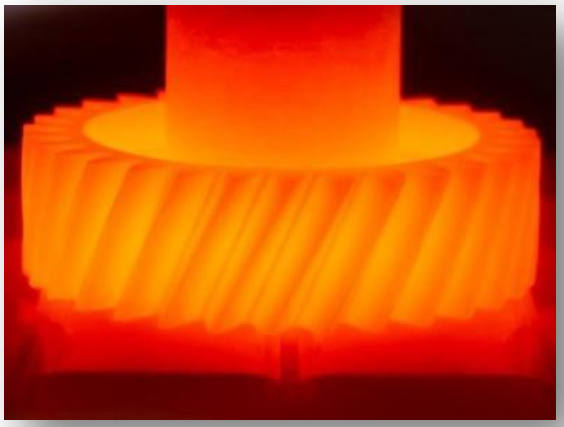

1
2
3
7.5
5
2.5
tau / GPa
25
50
75
100
125
150
Timestep

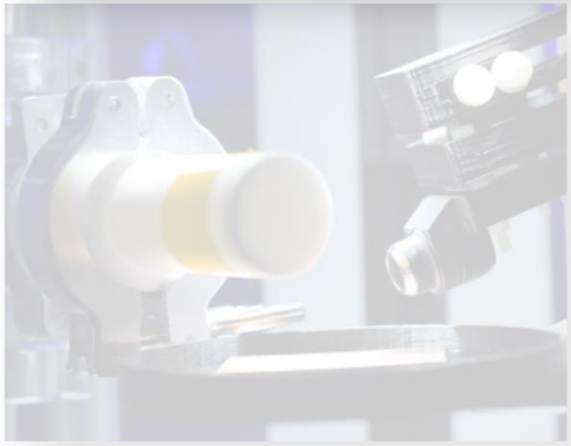


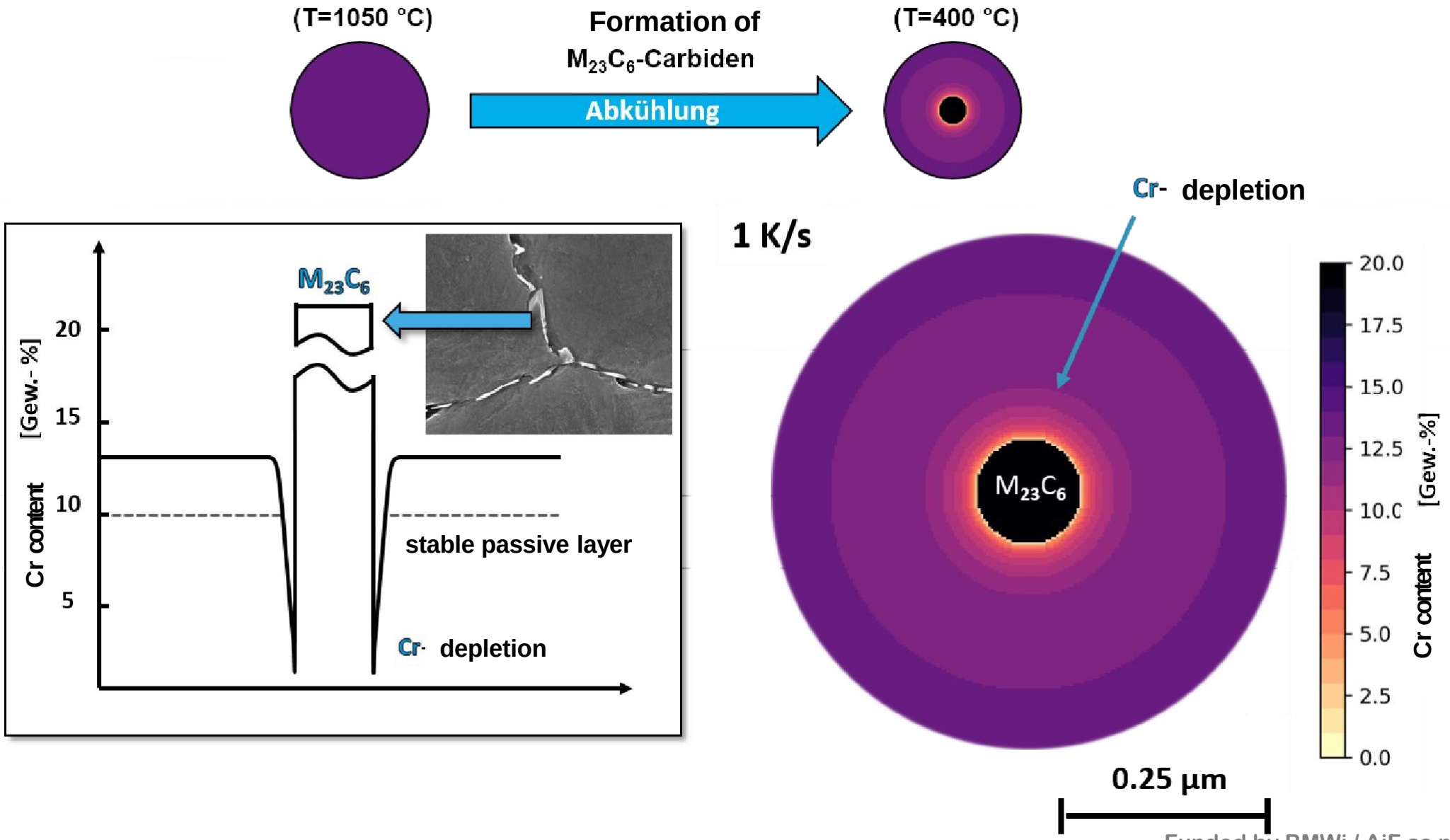

Temperatur
Austenitisierung
Erwärmen
Abschrecken
Anlassen
Temperatur
Austenitisierung
Erwärmen
Abschrecken
Anlassen

Heat treatment

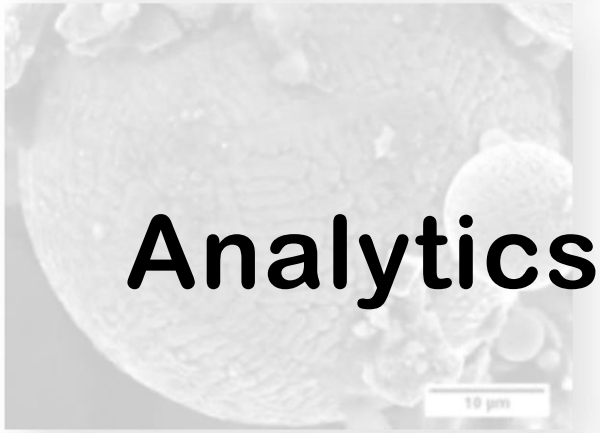


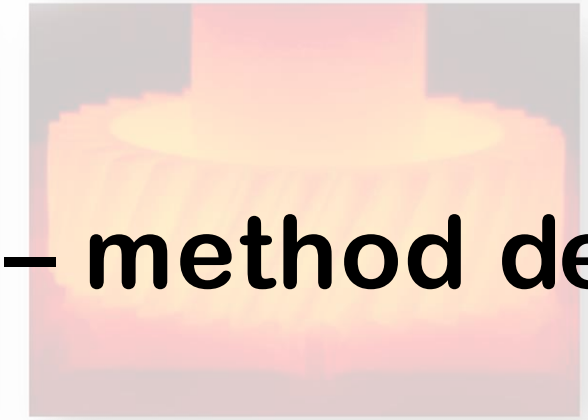


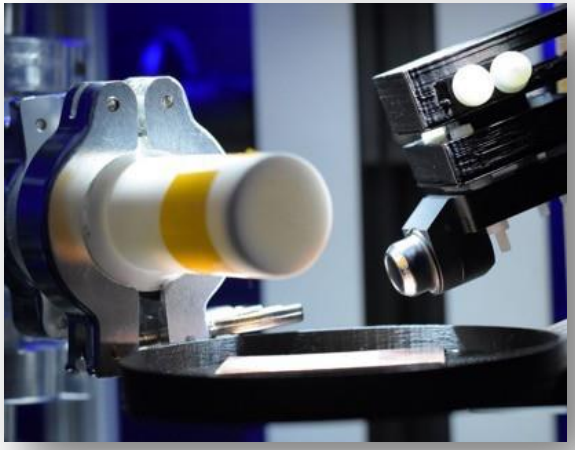


Funded by BMWi / AiF as project
„Anpassung der Wärmebehandlung martensitisch-nichtrostender Messerstähle zur Entwicklung spülmaschinenbeständiger Schneidwaren“

Analytics – method development



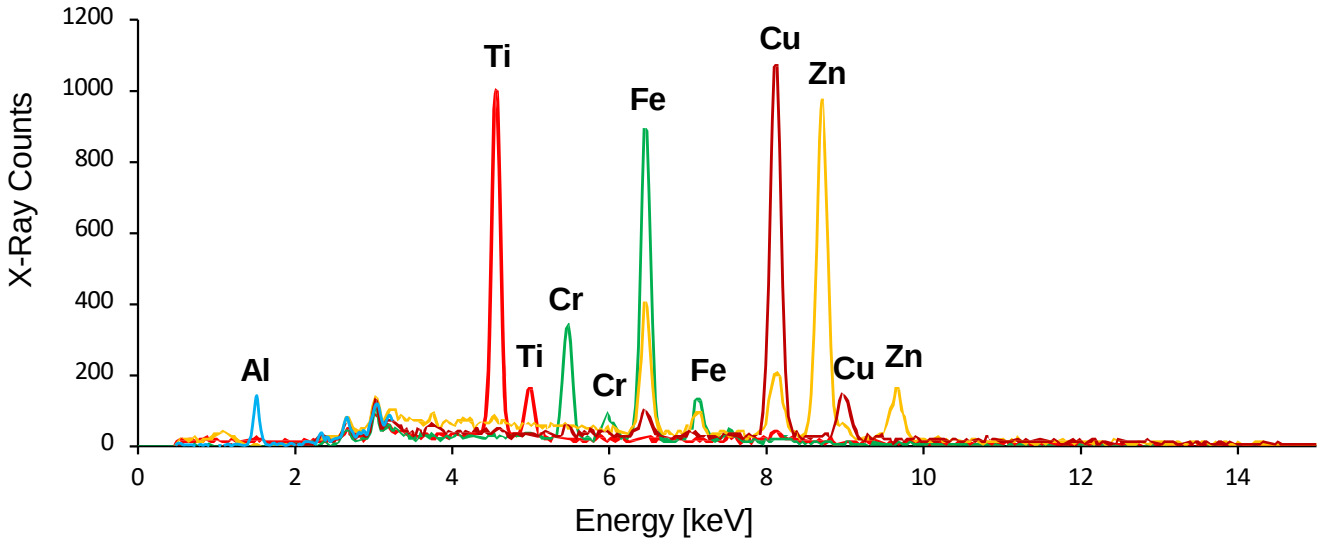


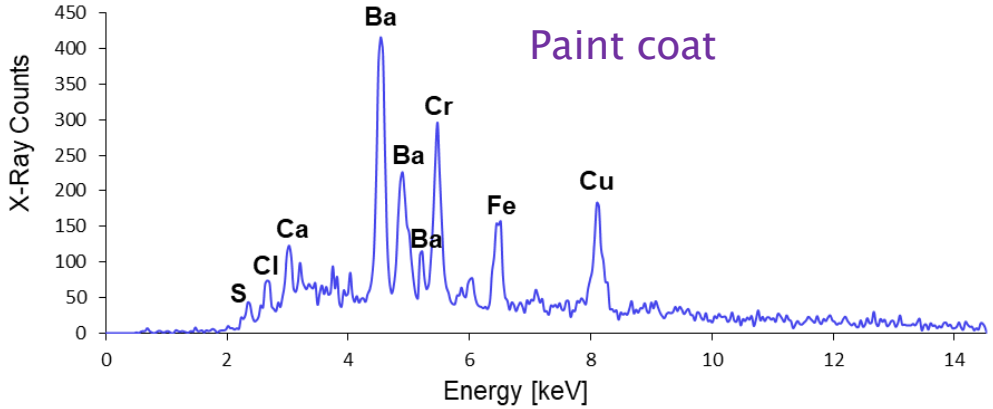



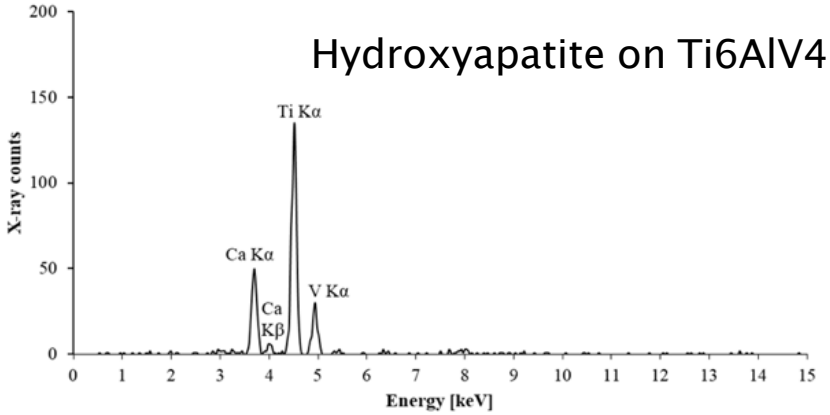
Chemical analyses performed directly on the light microscope



Titan - 1.4301 - galvanised sheet metal - Copper - Aluminium





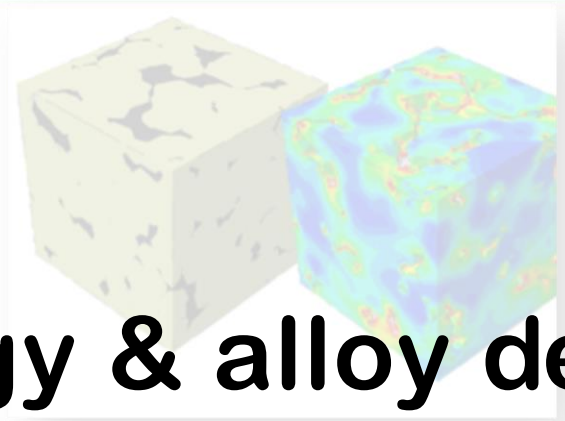


Use of X-ray fluorescence analysis

Funded by EU - EFRE Sachsen-Anhalt as project „X-ELMA: X-ray fluorescence elemental analysis for microscopic applications“

Metallurgy & alloy development



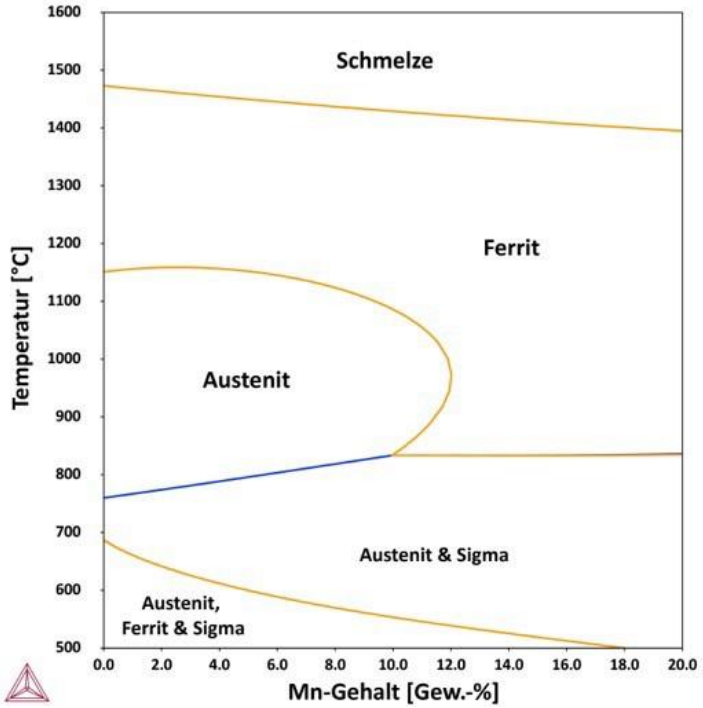


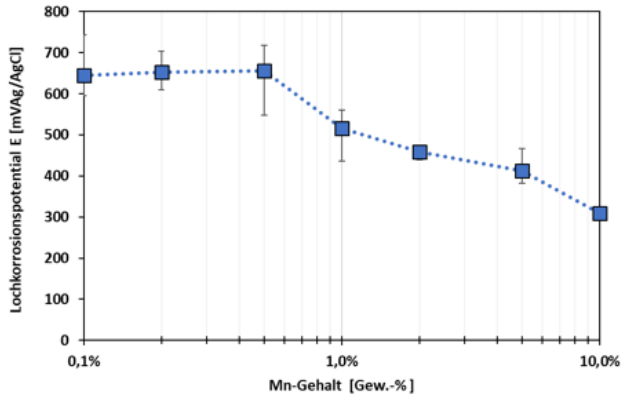


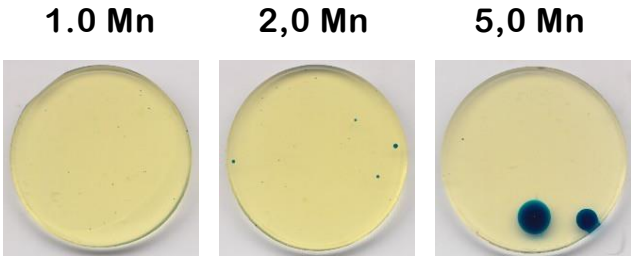


Alloy effects in stainless steels – e.g. Mn

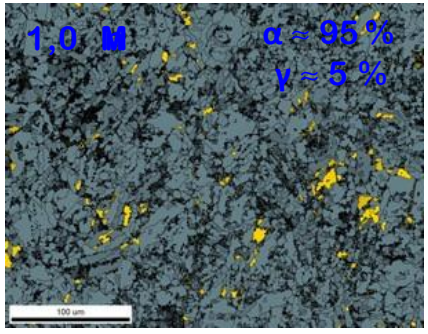


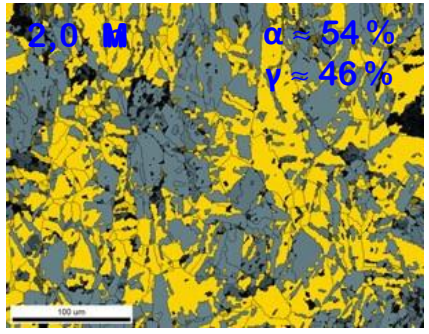


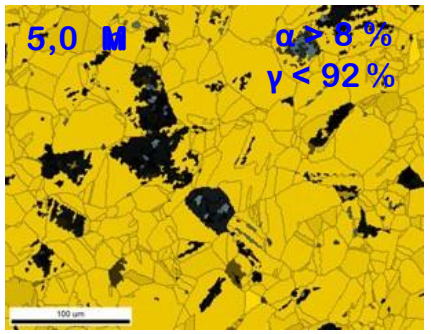












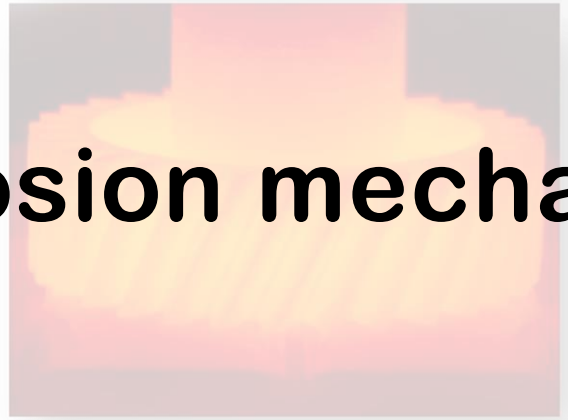
Objective : Experimental investigation of the influence of alloying and accompanying elements on the corrosion resistance of stainless steels

Partially supported by the German Research Foundation (DFG) in project 'Agar-based gel-electrolytes for corrosion diagnostic' (project number 330472124).

Corrosion mechanisms



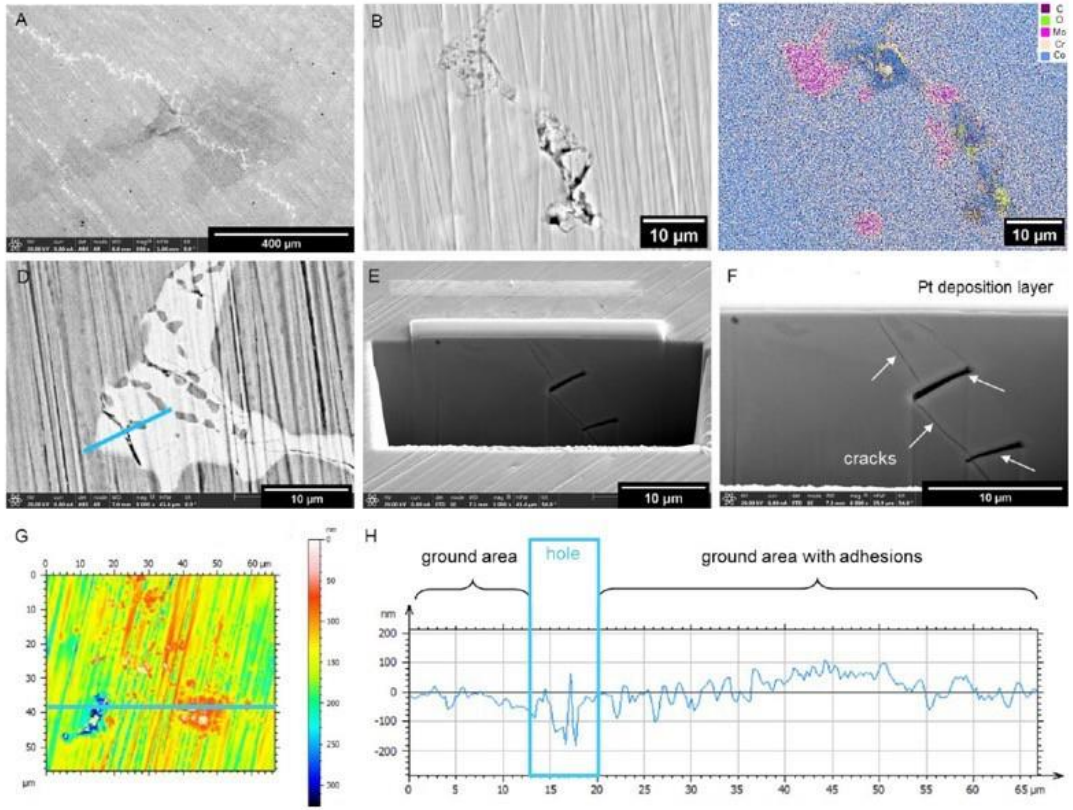


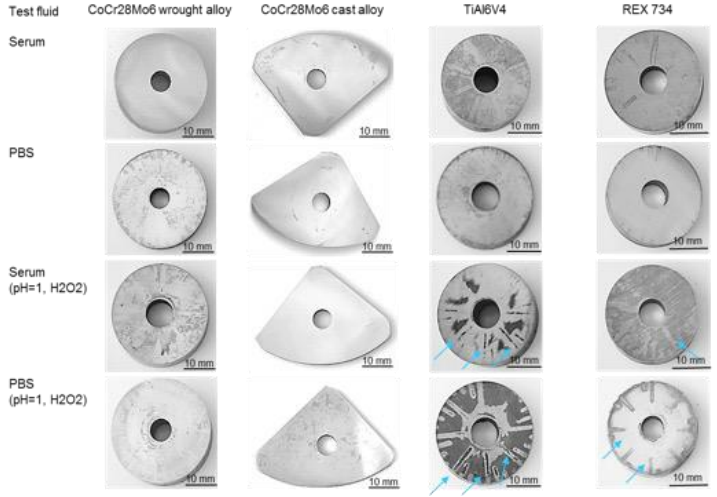




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Corrosion tests in medical technology





Crevice corrosion tests under simulated real conditions

Supported by the European Structural and Investment Funds (ESF) under the program "Sachsen-Anhalt WISSENSCHAFT Internationalisierung“, International Graduate School “MEMoRIAL” (project grant no. ZS/2016/08/80646).



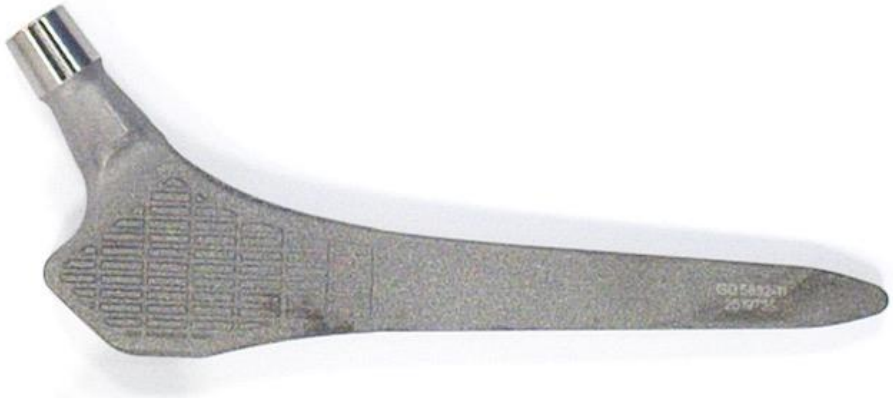




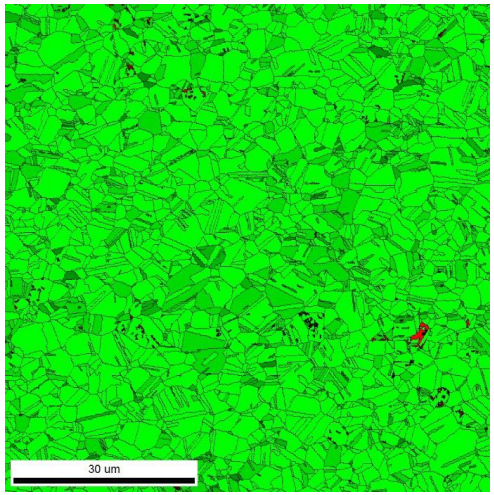


Medical technology – metallic prosthetics

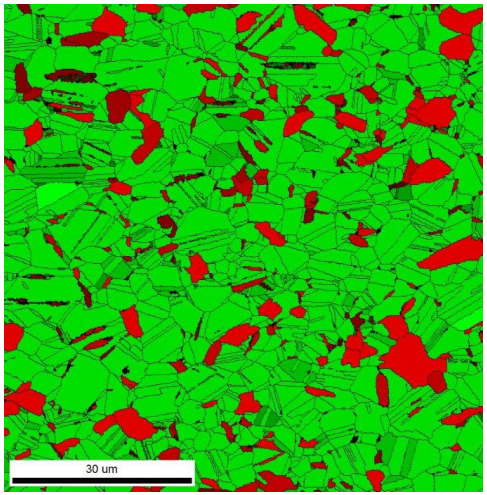
Phase transformation through heat treatment of CoCrMo



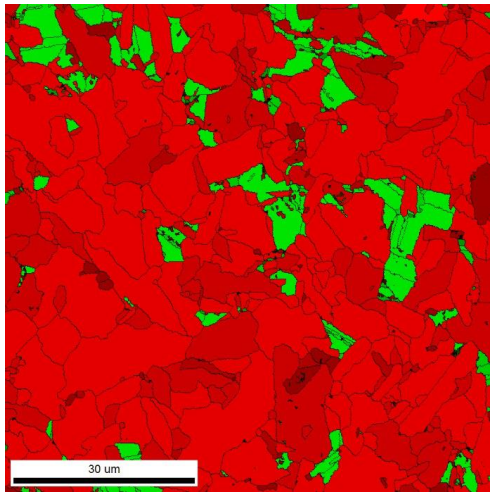
Initial state

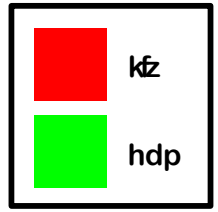


8 h HT (750 °C)



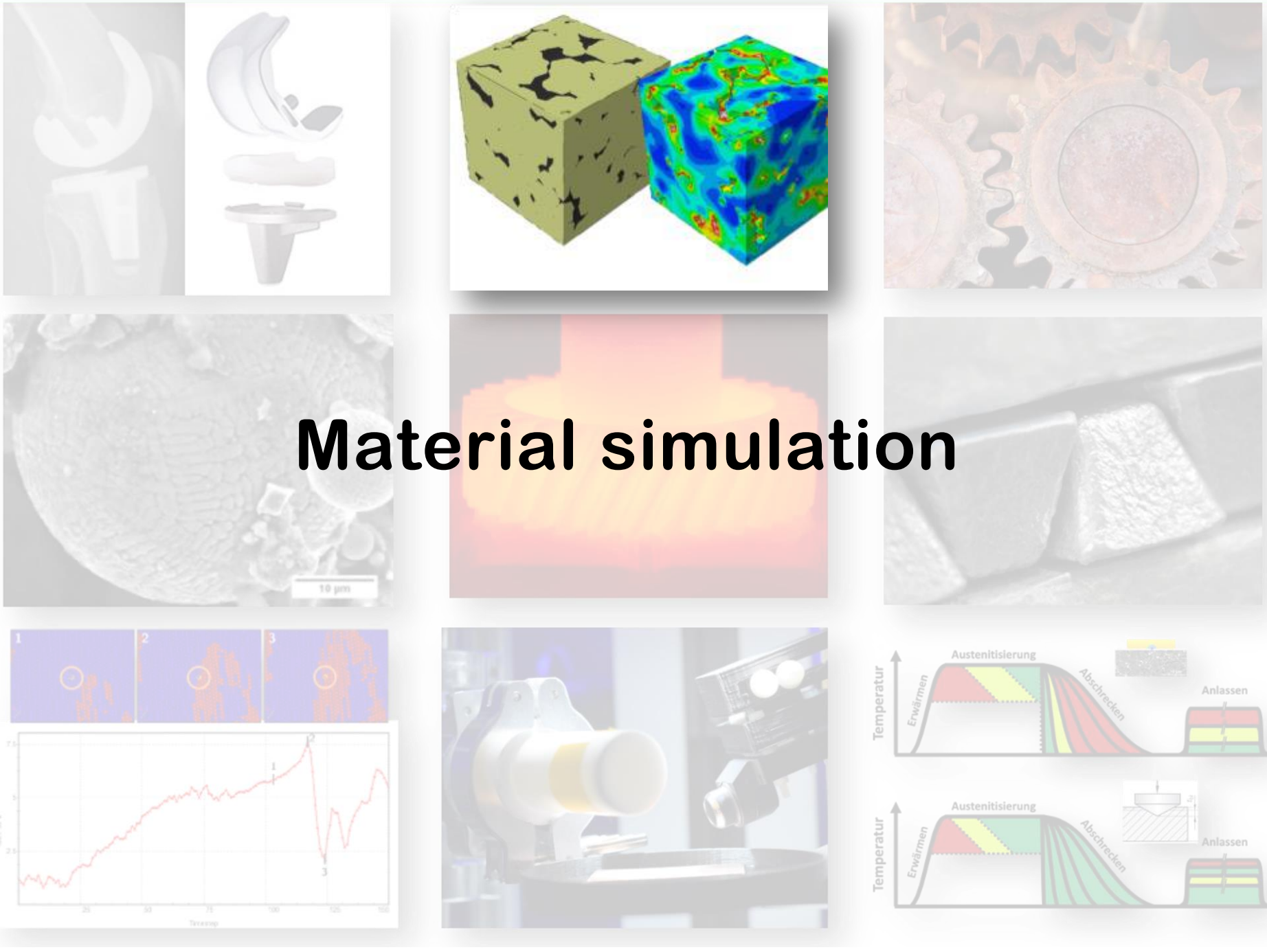
24 h HT (750 °C)





Forging alloy, automotive, small grain size

Supported by the European Structural and Investment Funds (ESF) under the program "Sachsen-Anhalt WISSENSCHAFT Internationalisierung“, International Graduate School “MEMoRIAL” (project grant no. ZS/2016/08/80646).

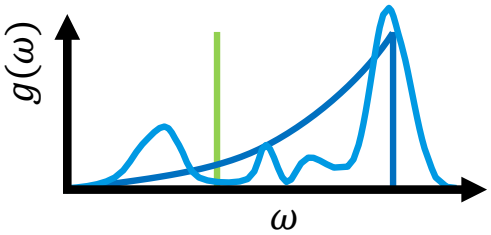


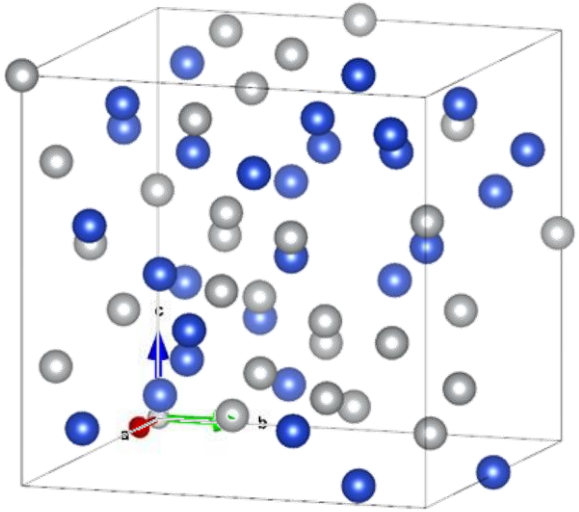
Material simulation

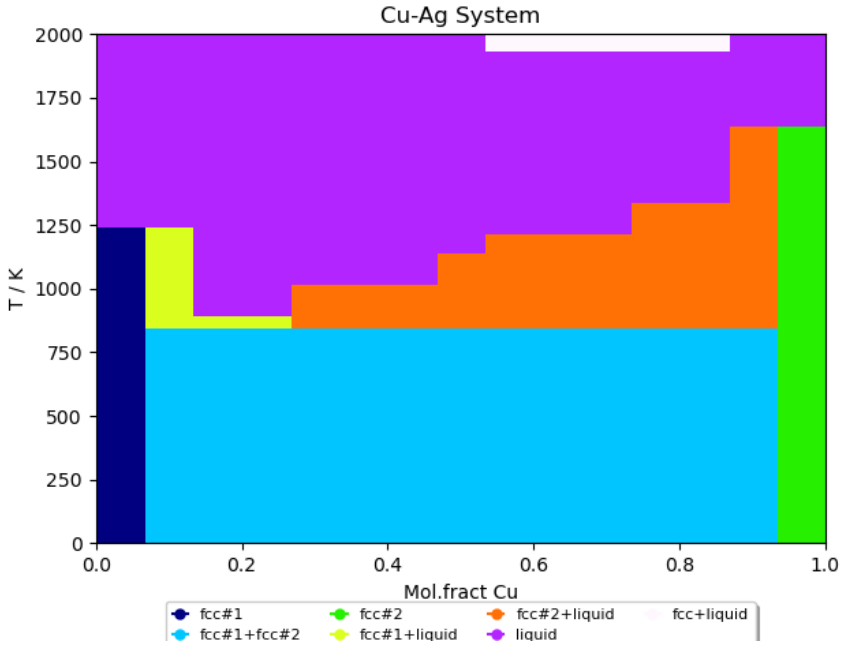
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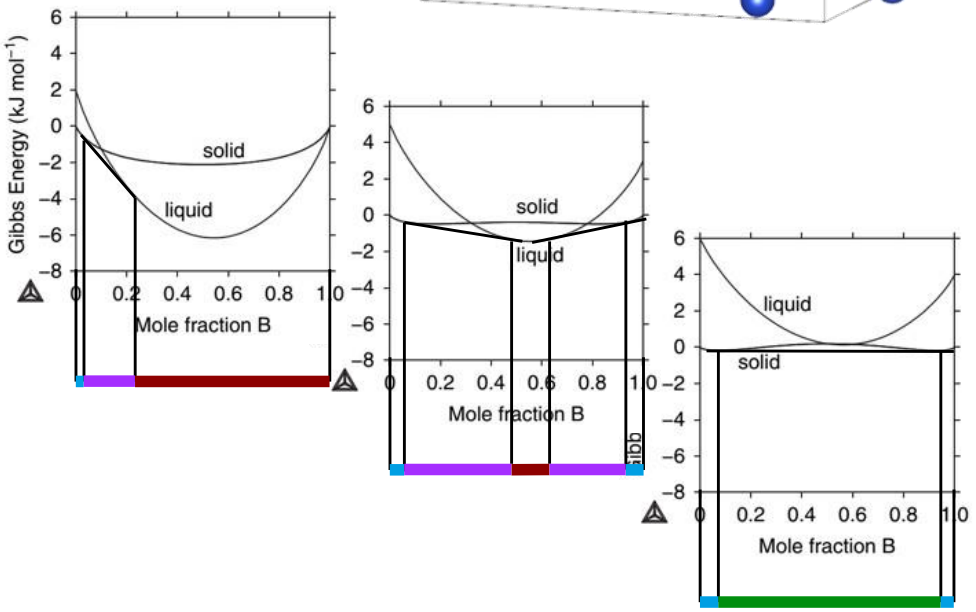
Calculation of phase diagrams from atomistic simulation

new method for calculating
phase diagrams without the need for time-consuming
empirical adjustments by using high-throughput
atomistic simulation





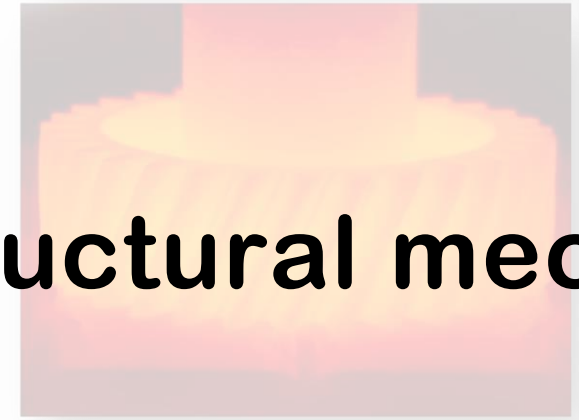




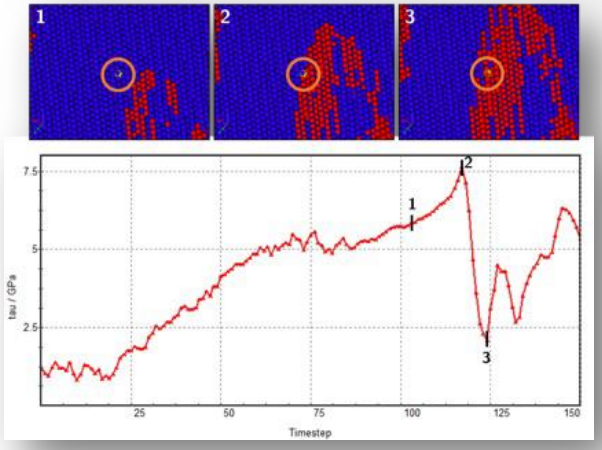
Partially supported by the German Research Foundation (DFG) in Graduate School (No. 1554) 'Micro-Macro-Interactions in Structured Media and Particle Systems.'

Microstructural mechanisms



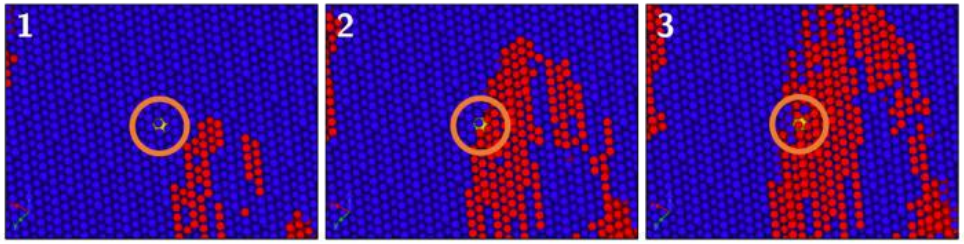






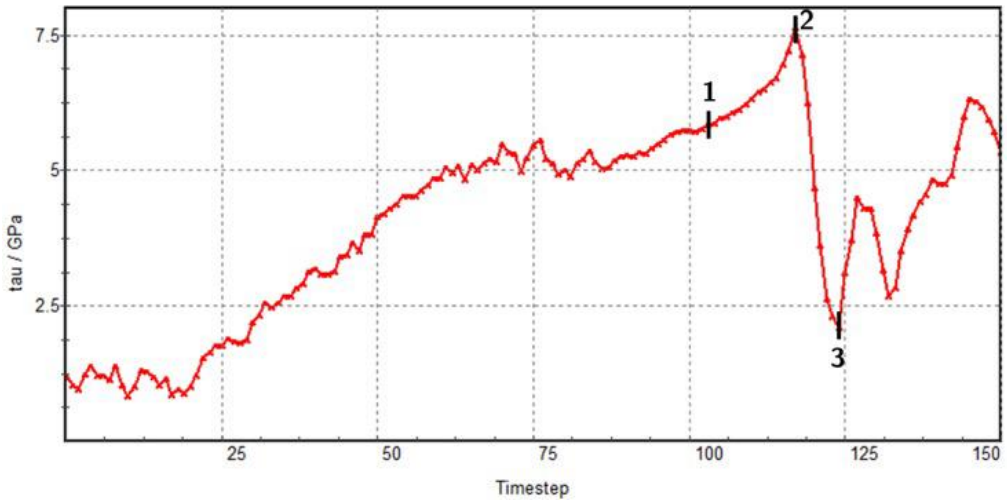
Twin formation in iron

Atomistic calculations / molecular dynamics on the formation mechanism

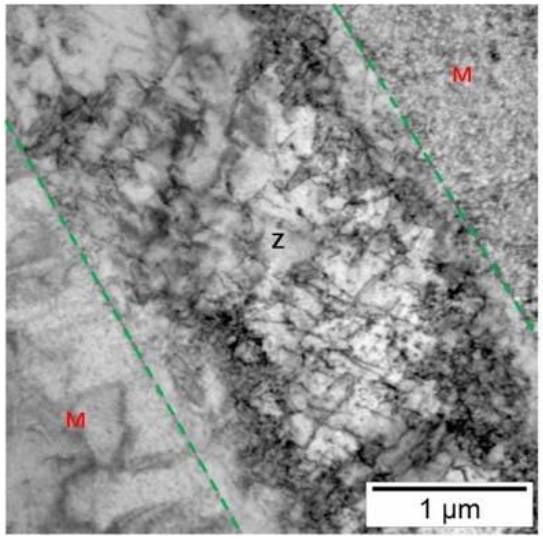


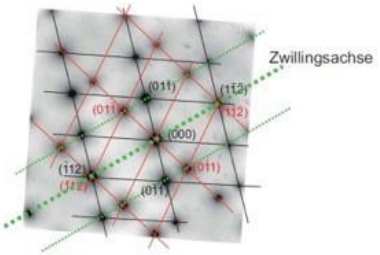
Matrix

Zwilling



TEM studies on dislocation activity / on the formation mechanism



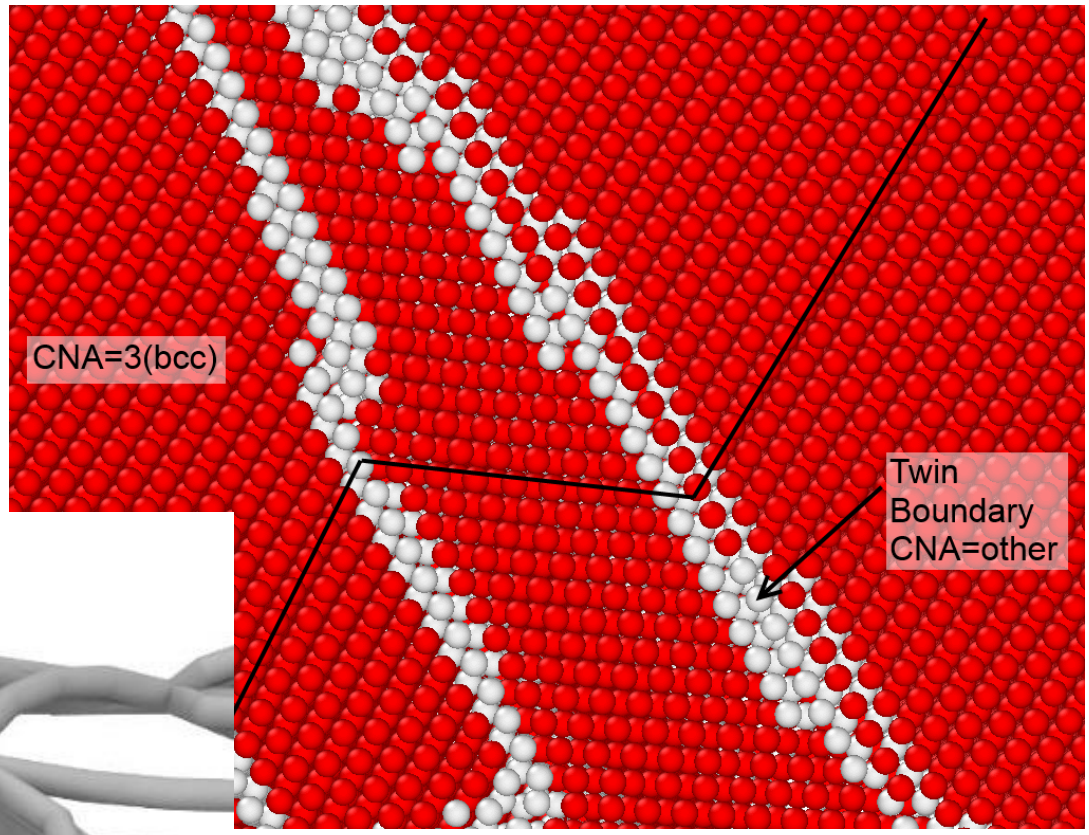


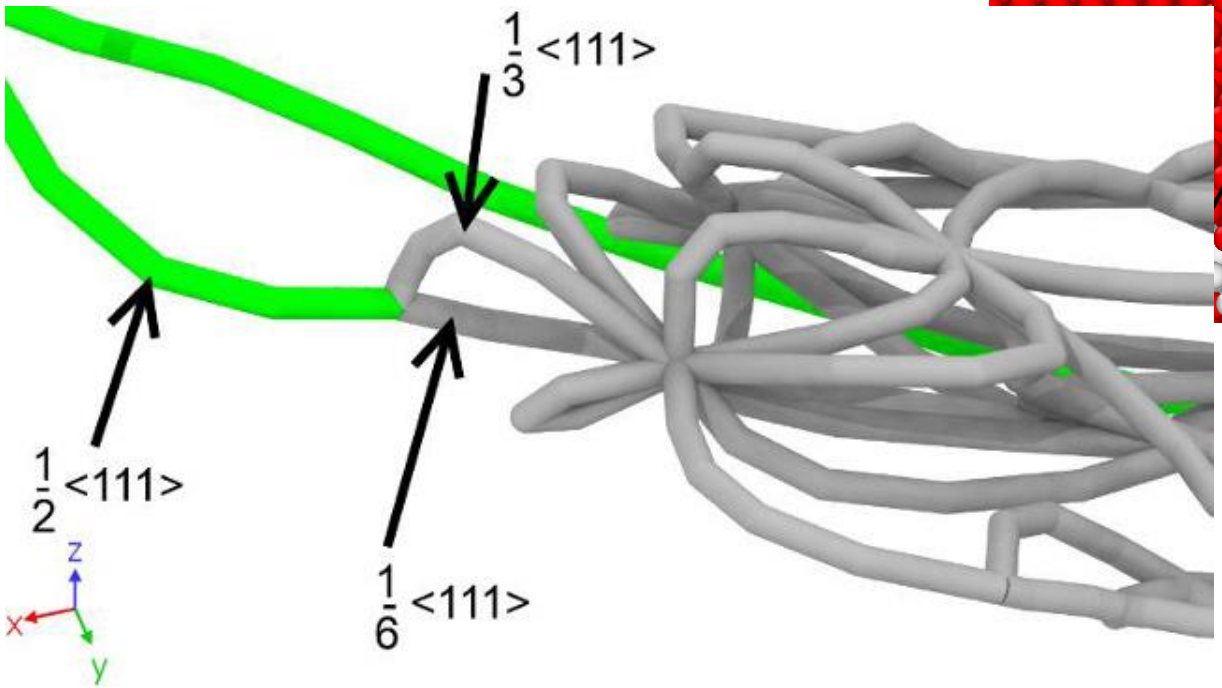
Partially supported by MDZWP e.V. The EBSD measurements were carried out by FEI SCIOS, funded by DFG large equipment funding (article 91b GG)

Twin formation in iron

Dislocation reaction as identified from atomistic simulation:

1/2[111] → 1/3[111] + 1/6[111]



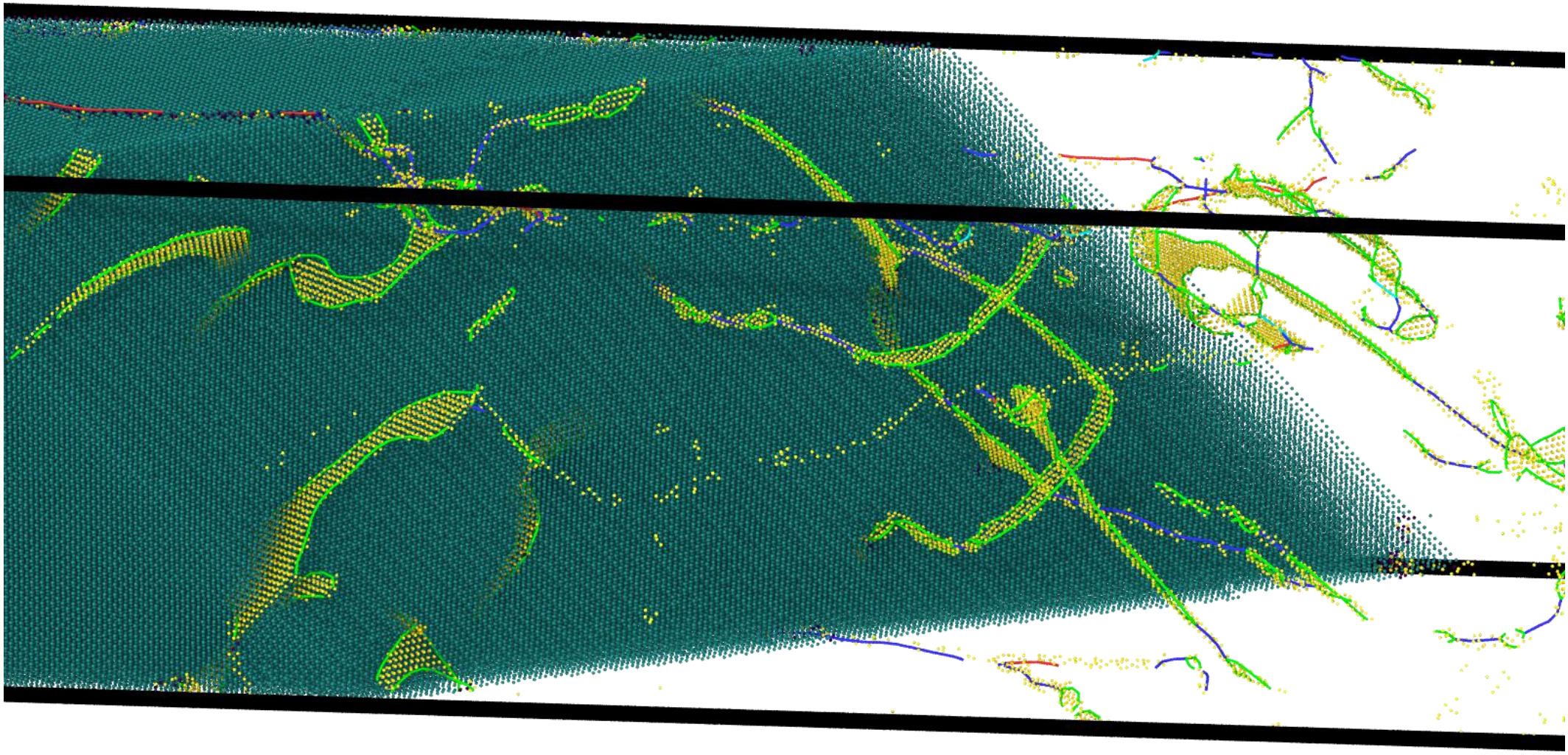


Partially supported by MDZWP e.V.
EBSD measurements were carried out by FEI SCIOS, funded by DFG large equipment funding (article 91b GG)

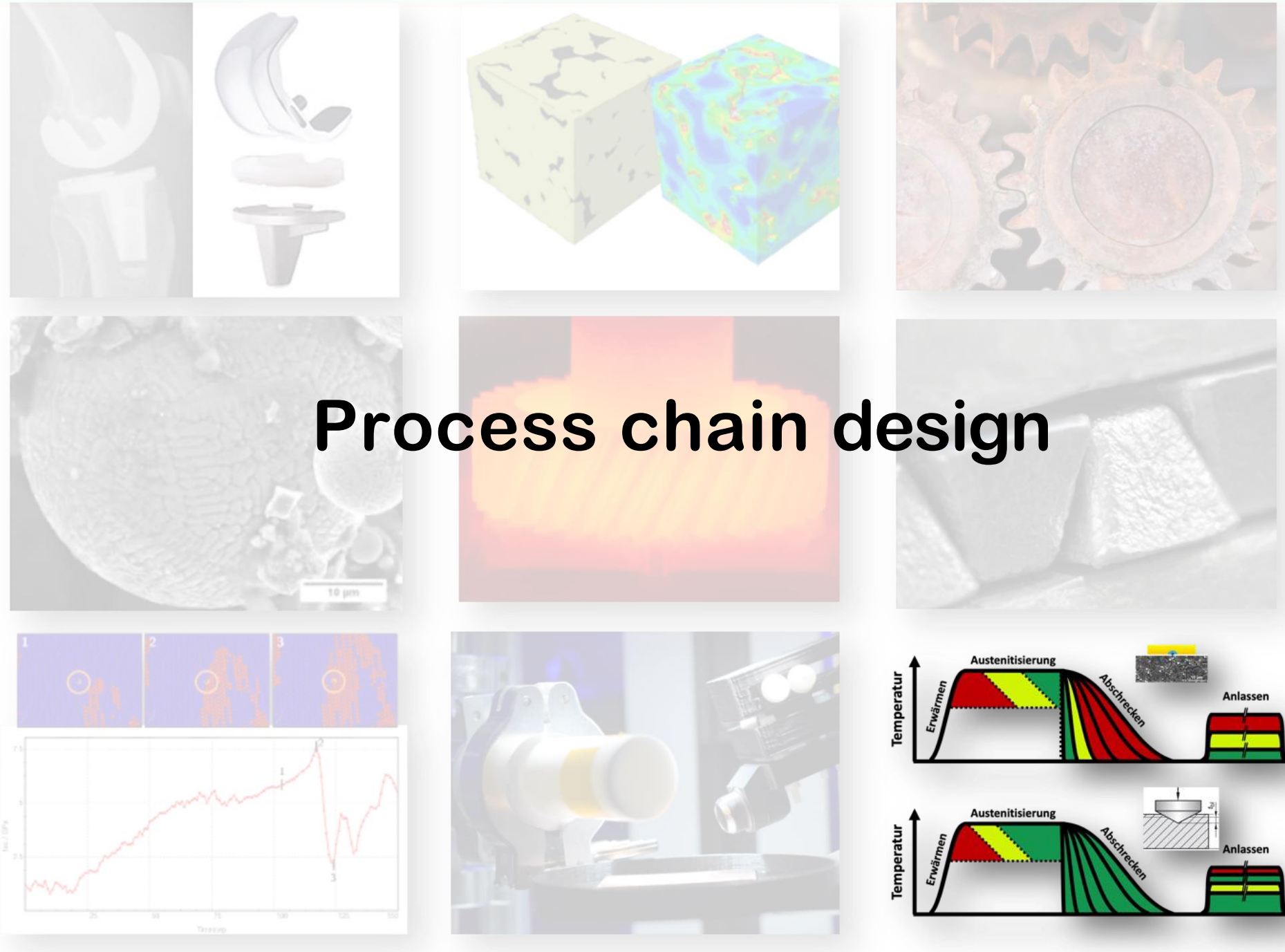
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Collision welding of Aluminium

High-velocity collision of metallic plates leads to welding
Investigation of the mechanisms and prerequisites

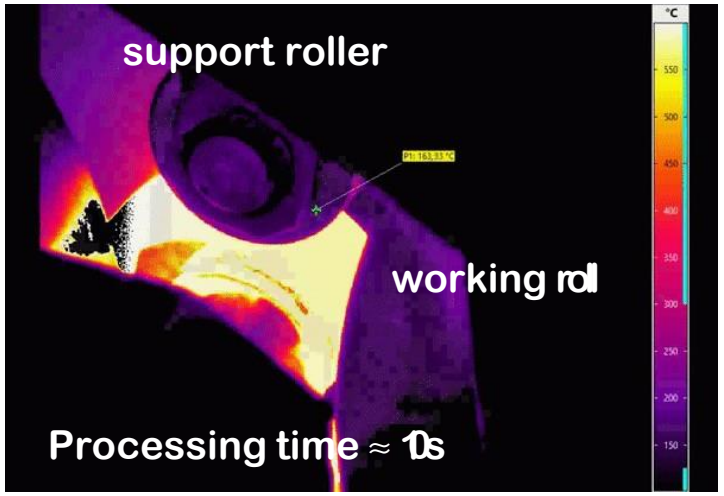


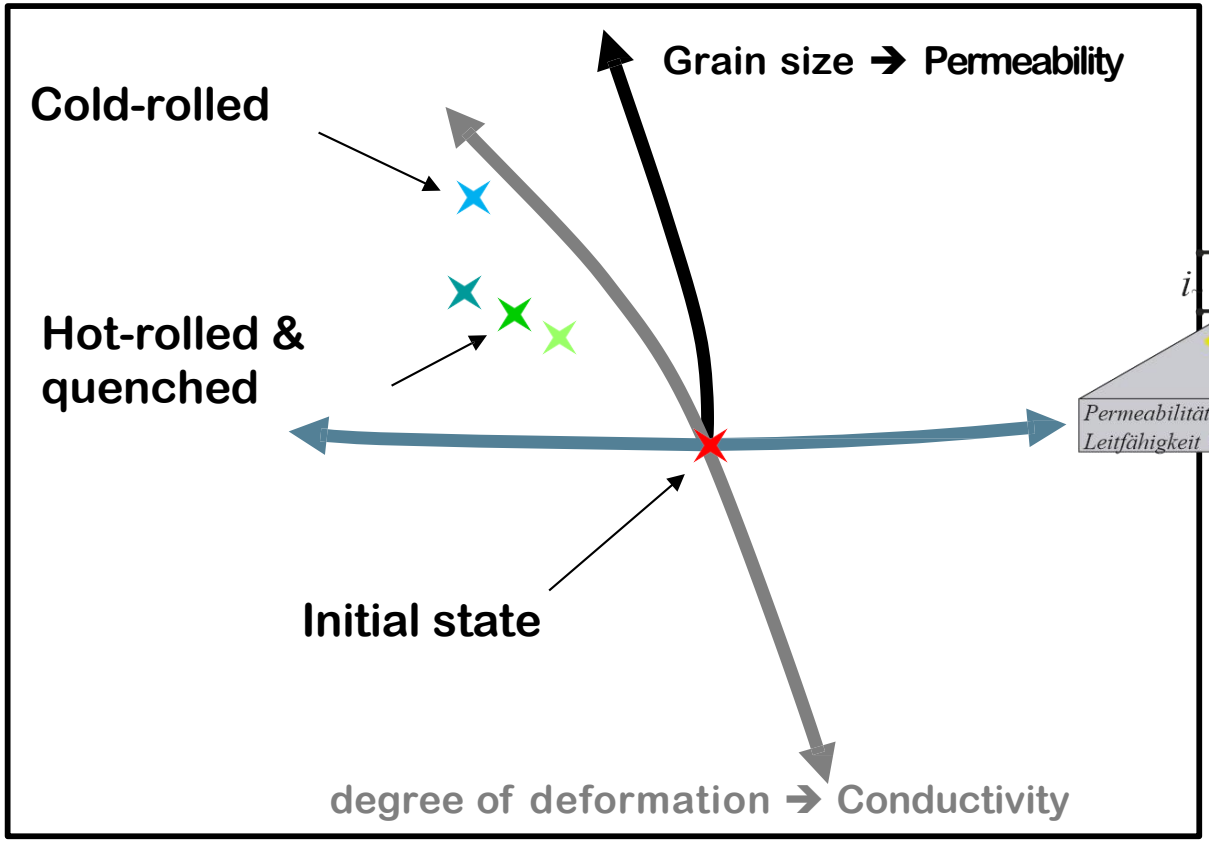
Process chain design

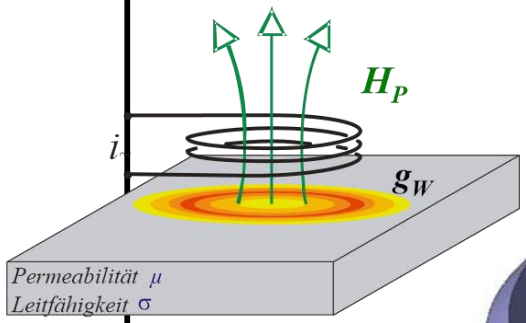


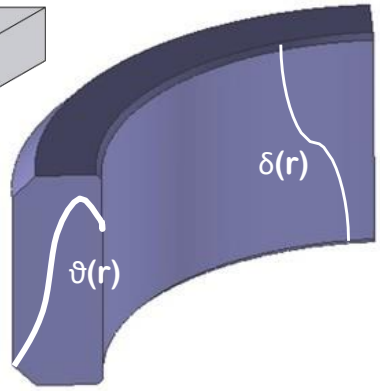
Thermomechanical ring rolling with predictive property control

Model-based control of a rolling process with the aim of adjusting certain surface layer properties through targeted and monitored model control using eddy current-based sensors







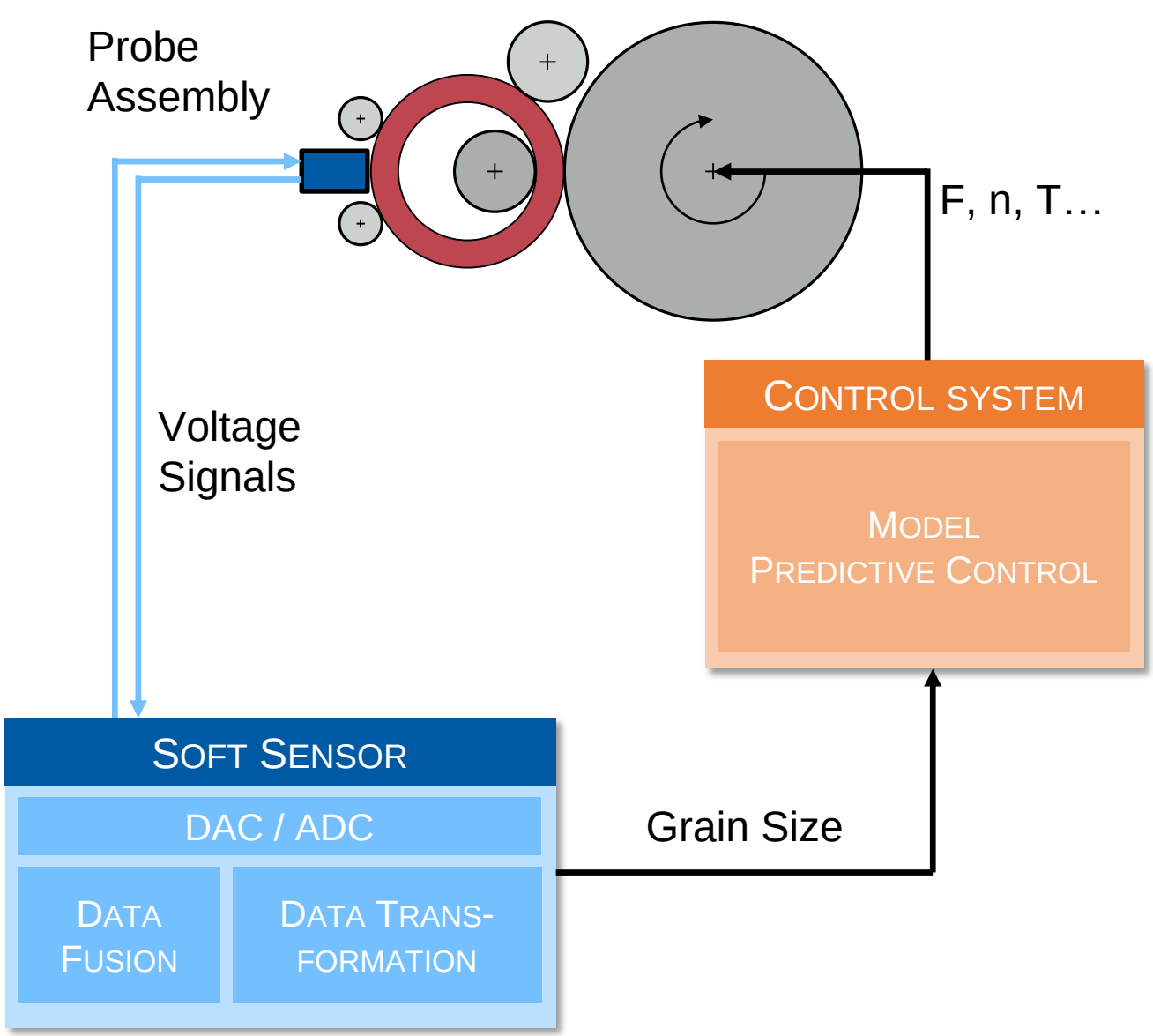


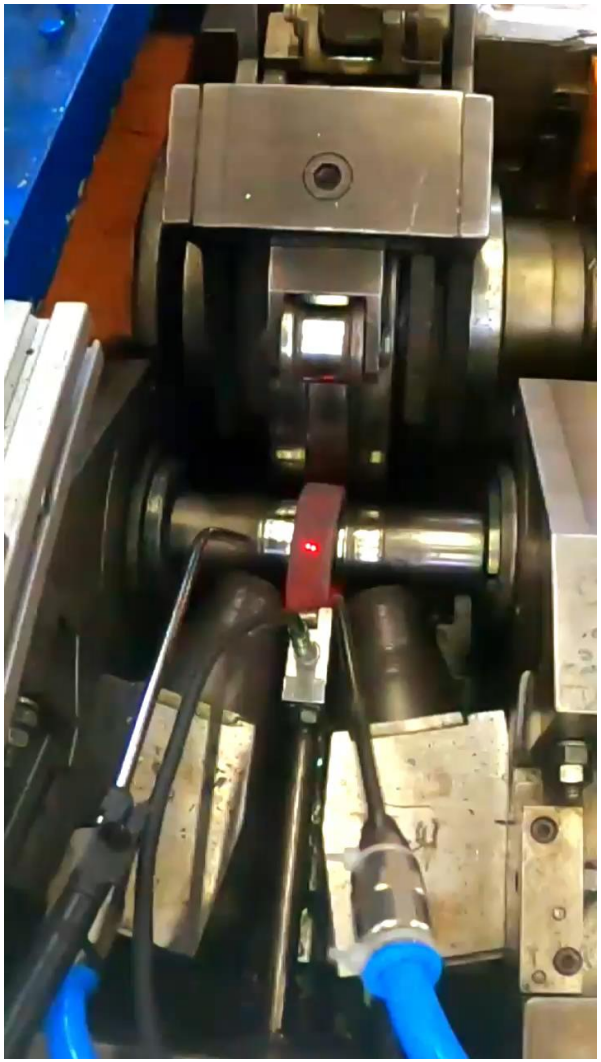
TECHNISCHE UNIVERSITÄT DRESDEN

FF Formgebende Fertigungsverfahren

SPP 2183

Thermomechanical ring rolling with predictive property control

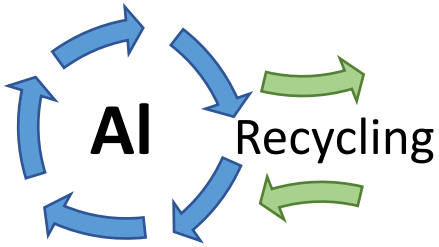




Supported by the German Research Foundation (DFG) in the priority program ‘SPP2183’ through project no. ‘424337466’.

Emmy-Noether junior research group

New alloy families made from recycled aluminum for sustainability and conservation of resources



https://gepris-extern.dfg.de/gepris/projekt/524220218?language=en



Laboratory of Primary Shaping Technology

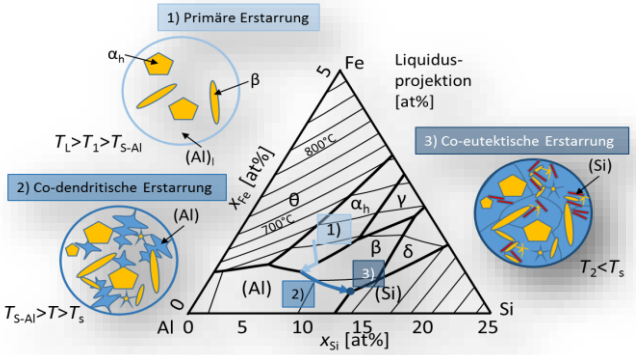
https://www.uut.ovgu.de/UT.html

- **Superiour understanding of process-structure-property interactions**
  - Recycling of Al alloys
  - **Joining of metals and alloys (diffusion, thermodynamics)**
  - **Processes of casting,** heat treatment**, joining** and additive manufacturing
  - High-throughput experiments
- **Characterization of materials properties**
  - **Complex crystal structures (intermetallic phases) and microstructures**
  - Mechanical properties

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Competencies and infrastructure at the IWTM

Phases, phase formation and microstructures

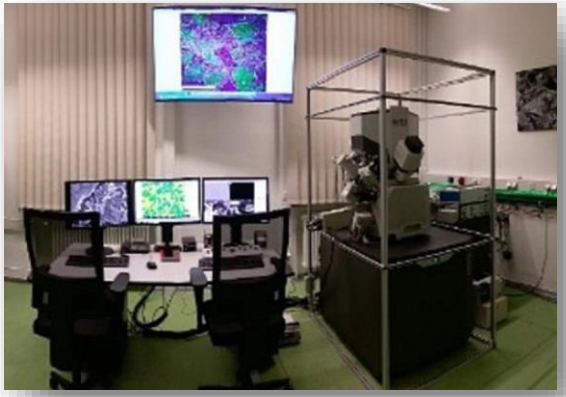


Effect of impurity elements – science of „dirty alloys“

Solidification and Heat treatments



Microstructure analysis SEM (EDX, EBSD), XRD, CT, …



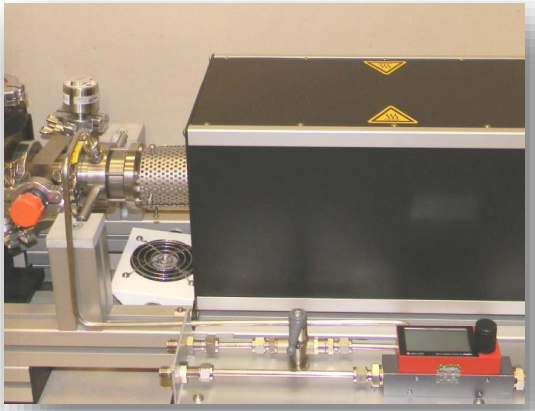
Alloy synthesis



Powder milling



Wetting behaviour



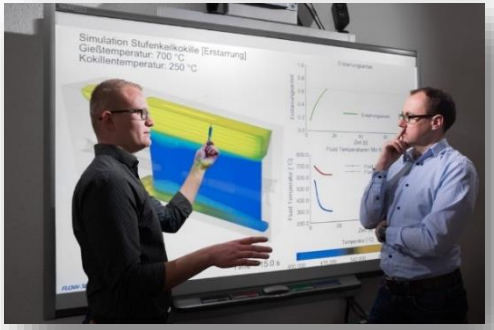
Mechanical testing



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Competencies and within the lab of primary shaping technology

Simulation



Gravitation casting Injection moulding



Additive manufacturing



Material characterization

